ATOMARINE

PRE-SEED · 2026 · CONFIDENTIAL

Nuclear Propulsion as a Service

Decarbonizing global shipping with detachable offshore nuclear power.

DIMITRIS KOUTENTAKIS · EMILE GERMONPRÉ

THE STAKES

Bunker fuel's economics are getting worse

Slow-steaming is a temporary workaround

2-3x FUEL PRICE CHANGES IN MONTHS

Volatility is the main OPEX risk.

>30% SPEED REDUCTION

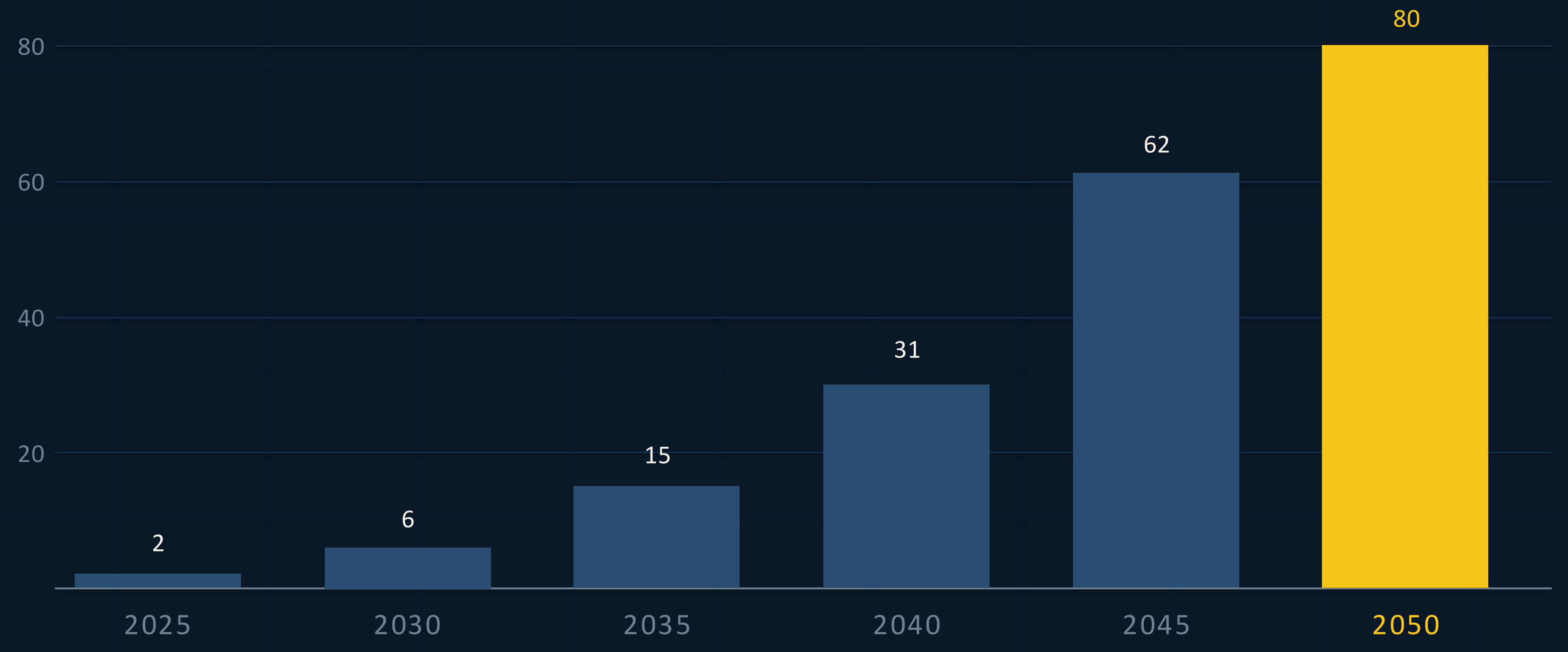
Vessels operating below design speed.

40× FUELEU PENALTY RAMP BY 2050

Penalty intensity from baseline year.

FUELEU MARITIME PENALTY INTENSITY

% REDUCTION REQUIRED VS 2020



SOURCES · CLARKSONS · IEA · IMO GHG STUDY 4 · FUELEU MARITIME REGULATION (EU 2023/1805)

THE ONLY PATH

Nuclear checks every box. Nothing else does.

NUCLEAR

The only proven option.

- ✓ Scalable supply chain
- ✓ Proven: 60+ years at sea
- ✓ Cost-effective at scale
- ✓ 100% zero-emission
- ✓ Maintains full speed

But conventional nuclear shipping has been blocked for 60 years.

✗ Ammonia / H_2

No scalable supply chain. Unproven at scale. 3–5× fuel cost.

✗ Wind / Sails

Supplemental only. Cannot propel deep-sea vessels independently.

✗ Biofuels

Land-use competition. Limited supply. Not fully carbon-neutral.

✗ Carbon Capture

Unproven at maritime scale. Adds weight, cost, and complexity.

THE PARADOX

Non-technical barriers have blocked nuclear shipping for 60 years.

01

Denied Access

Nuclear ships denied entry to commercial ports worldwide. Emergency zones around population centers make it impossible.

02

Slow regulation

Every state demands separate licensing. Onshore reactor licensing can take a decade.

Infeasible

03

High cost

Diseconomies of scale due to low power and low utilization and billion-dollar liability on shipowners balance sheet

04

Liability

Billions in nuclear liability per vessel. No shipowner will take this on their balance sheet.

Unattractive

ARCHITECTURE

Atomarine separates the reactor.
The LeadShip powers the ocean leg.

01 **Depart**
Ship leaves port under its own power.

02 **Rendezvous**
Meets LeadShip at a waypoint, beyond International Waters.

03 **Connect**
Dynamic ship-to-ship cable

04 **Transit**
99% of voyage under nuclear power · 15+ kn.

05 **Disconnect**
Before territorial waters; ship enters port alone.

LEADSHIP + CONVOY · DYNAMIC HV CABLE

COMPETITIVE EDGE

One insight → Every barrier resolved

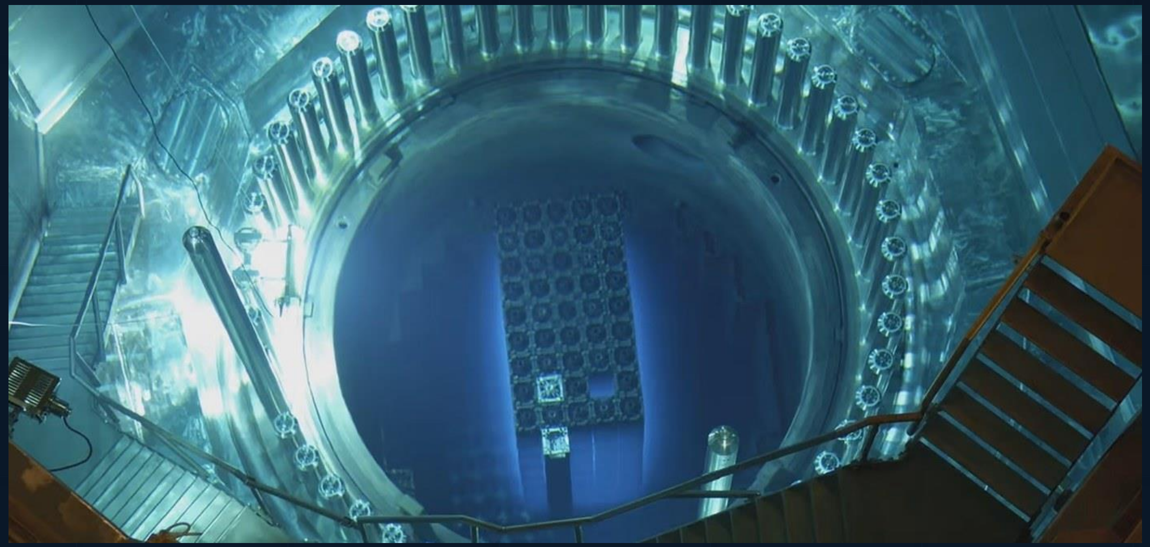
	BARRIER		ATOMARINE RESOLUTION
✗	Port access denied	→	✓ LeadShips never enter port — international waters only.
✗	Liability & insurance	→	✓ Zero nuclear liability on shipowner balance sheets.
✗	Multi-jurisdiction regs	→	✓ Single Flag State licensing — one design, one jurisdiction.
✗	Public opposition	→	✓ No reactors near population centers.

Every competitor puts reactors on cargo ships. We removed the reactor from the ship entirely.

TECHNOLOGY

Technology Proven for Decades

PILLAR 01

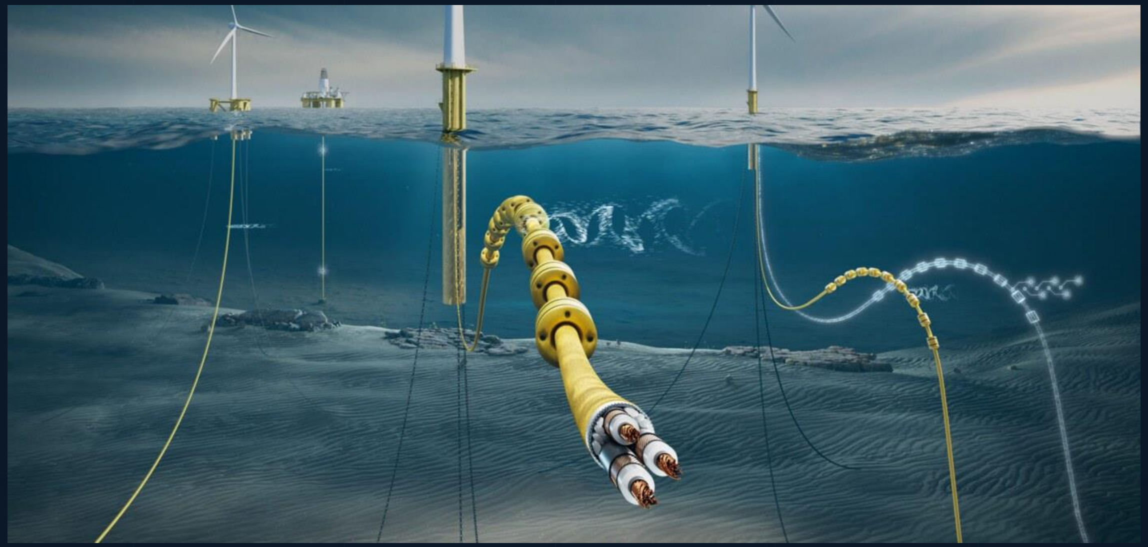


Marine iPWR Reactors

160+ NAVAL VESSELS · 2,000+ REACTOR-YEARS

Integral PWR with 70+ years of maritime heritage. Modern SMR designs already in licensing.

PILLAR 02

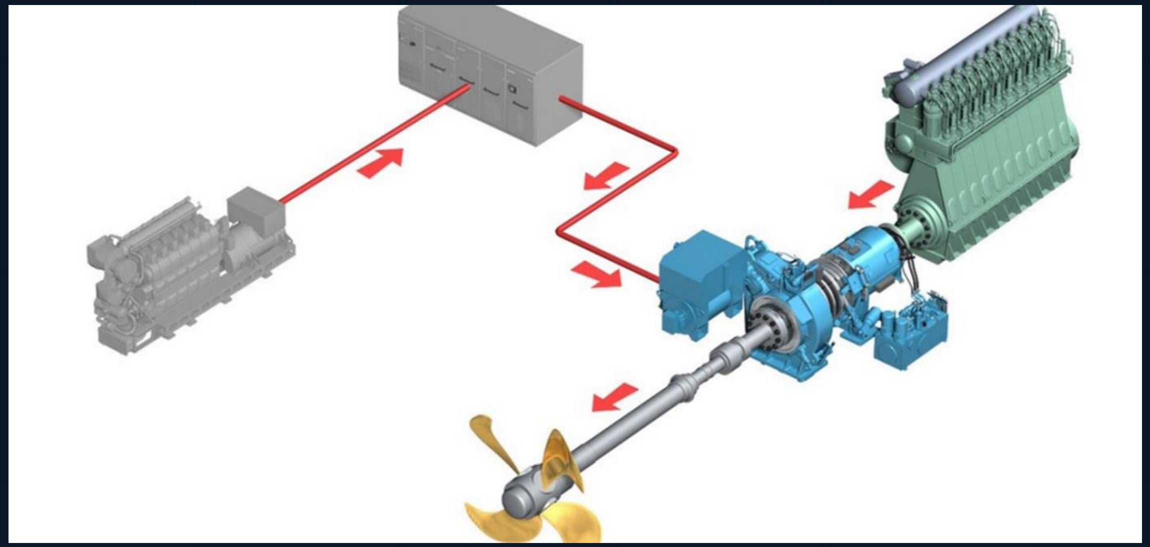


Dynamic Offshore Cables

PROVEN IN FLOATING WIND & OIL/GAS

High-voltage dynamic power cables rated for open-ocean conditions. Multiple commercial manufacturers.

PILLAR 03



Electric-Drive Propulsion

THOUSANDS OF SHIPS ALREADY EQUIPPED

PTO / PTI systems standard in LNG carriers, cruise ships, icebreakers. Retrofit cost: $2–3M.

BUSINESS MODEL

Nuclear propulsion as a service

REVENUE MODEL

PRICING	Per-voyage fee indexed to energy + avoided carbon penalties.
CONTRACTS	Long-term take-or-pay, secured by EU carbon-tax savings.
REVENUE	$150–250M annually per LeadShip at >90% utilization.
MARGINS	Staffing doesn't scale with power; convoys unlock strong margins.

WHY SHIP OPERATORS BUY

- ✓ Total cost < fuel + carbon tax
- ✓ Zero nuclear liability on shipowner balance sheet
- ✓ No specialized crew needed
- ✓ Restore full sailing speed
- ✓ Compliance credits banked & sold
- ✓ Minimal retrofit (~$2–3M PTI)
- ✓ Maintain fuel optionality (HFO/H_2/Ammonia)

UNIT ECONOMICS

Convoys unlock economies of scale

$1.0–1.6B

CAPEX

Hull + reactor + NRE.

60 yr

DESIGN LIFE

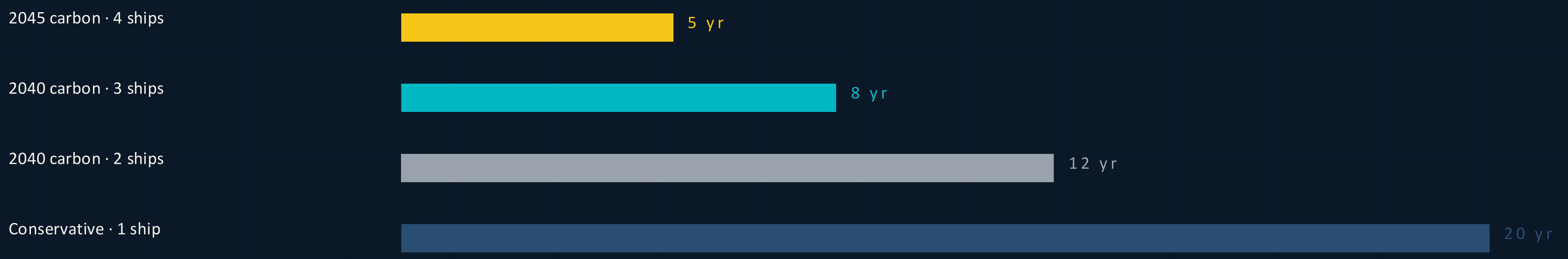
2× merchant ship lifespan.

$150–250M

ANNUAL REVENUE

Per LeadShip at >90% utilization.

PAYBACK BY SCENARIO



22% LEARNING RATE · AT FLEET SCALE (50 LEADSHIPS) PER-UNIT CAPEX DROPS BY >60%

MARKET

A $127B+ market under existential pressure.

TAM

$127B+

ANNUAL GLOBAL SHIPPING FUEL SPEND

Plus $1–100B in FuelEU carbon costs.

SAM

$95B

BULK + TANKERS · ROUTES > 1,000 NM

75% of fuel consumption.

SOM

$1.5B+

CONTAINER SHIPS · EU-REGULATED ROUTES

First wedge into the market.

BEYOND MARITIME

- **Floating datacenters**

  Centrally build self-powered datacenters and ship them to cool locations.

- **Remote energy supply**

  Plug in to any coastal grid; provide carbon-free energy to desalination plants.

- **Industrial decarbonization**

  Carbon-free power for hard-to-abate offshore loads.

- **Defense & logistics**

  Charge electric autonomous vessels; power deployments and emergency response.

MOAT

Once scaled, hard to displace.

01

Regulatory lock-in

First Flag State nuclear code creates path dependency.

02

Network effect

More LeadShips → shorter wait times → better service.

03

Technology lock-in

Ships designed around our connectors. High switching costs.

04

Intellectual property

Provisional patents filed. Multi-jurisdiction filings planned.

05

Talent moat

First to assemble a nuclear-maritime cross-disciplinary team.

TEAM

Expertise and market access.

Dimitris Koutentakis

CO-FOUNDER & CEO

MIT MBA · Naval Engineering MS

MIT EECS Undergrad · MEng

Amazon electrification optimization

Greek shipping background

Emile Germonpré

CO-FOUNDER & CTO

MIT Nuclear Science & Engineering PhD

Nuclear reactor systems specialist

UGent Chemical Eng. · Valedictorian

Fulbright Scholar

KEY PARTNERS & AFFILIATIONS

MIT	MPC CONTAINERS	COSTAMARE	DORIAN LPG	EXMAR	X-ENERGY	APOLLO ATOMICS	THE ENGINE

VISION

The fuel layer of shipping.
Then everything that moves on water.

HORIZON 1

TODAY → 2030

Container convoys

LeadShips decarbonize major trans-ocean container routes where ETS exposure bites first.

HORIZON 2

2030 → 2035

Autonomous barges, electric tugs

Ports adopt shore-side electric tugs; autonomous feeder barges pull power from offshore LeadShips.

HORIZON 3

2035 +

Mobile nuclear utility

Offshore compute, naval logistics, disaster response, remote industrial loads — a new category of utility.

The fuel layer of shipping becomes the electricity grid for everything mobile at sea.

TRACTION

Motion on three axes: commercial, technical, regulatory.

COMMERCIAL

Four data-exchange partnerships active.

Costamare
DATA-EXCHANGE

Dorian LPG
DATA-EXCHANGE

MPCC
DATA-EXCHANGE

Exmar
DATA-EXCHANGE

TEA v4
BREAK-EVEN AT N = 4

LOIs
TARGETED NEXT 6 MONTHS

TECHNICAL

Reactor vendor + both classification bodies.

X-Energy
LETTER OF SUPPORT

DNV
CLASSIFICATION ENGAGEMENT

ABS
CLASSIFICATION ENGAGEMENT

MIT Sandbox
NON-DILUTIVE FUNDING

Convoy GPS
GLOBAL POSITIONS MODEL

MIT delta v
FOUNDER TO CONFIRM

REGULATORY

Flag State framework + IP filed.

USPTO
PROVISIONAL PATENTS FILED

Flag State
SHORTLIST · FRAMEWORK

Maritime law
COUNSEL ENGAGED

Data-exchange partnerships are active; LOIs are targeted within 6 months — not signed today.

ROADMAP

Staged de-risking to first commercial LeadShip.

YEAR 0–1

PRE-SEED

$2–5M

— TEA & feasibility
— Regulatory scoping & NTQ
— Secure LOIs
— Advisory board

YEAR 1–3

SEED

$10–15M

— FEED engineering
— AiP from ABS
— Vendor selection
— Cable demo at sea

YEAR 3–8

SERIES A&B

~$250M

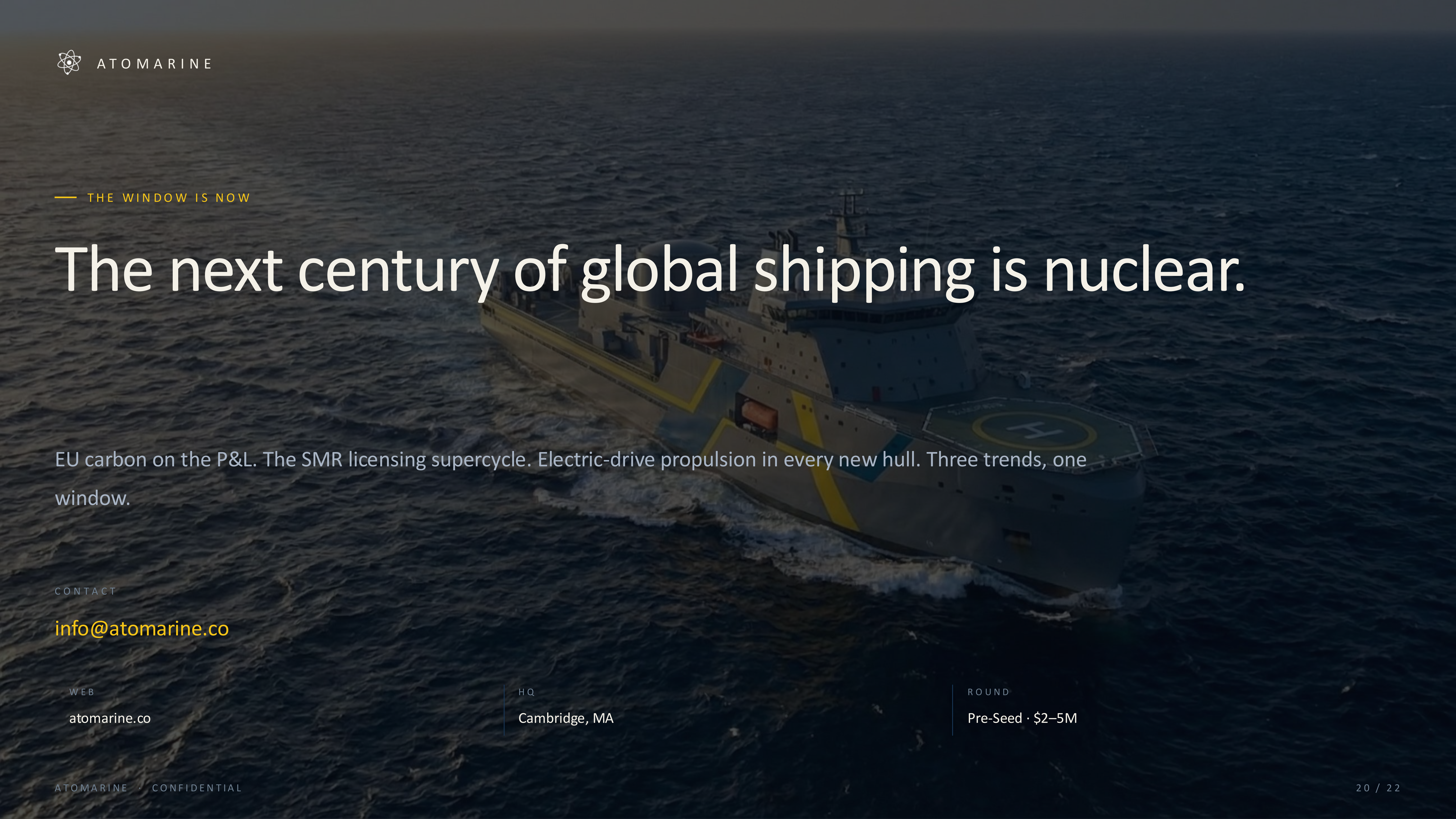
— Detailed engineering
— Full design approval
— Prototype hull
— Project financing

YEAR 8–15

SERIES C+

$1–10B

— FOAK construction
— Fleet ramp-up
— Production scale
— Profitability



THE WINDOW IS NOW

The next century of global shipping is nuclear.

EU carbon on the P&L. The SMR licensing supercycle. Electric-drive propulsion in every new hull. Three trends, one window.

CONTACT

info@atomarine.co

WEB	HQ	ROUND
atomarine.co	Cambridge, MA	Pre-Seed · $2–5M

WHY NOW

The Post-Combustion Era starts now.

Three independent forces. One unmissable alignment.

01

Regulatory certainty

EU ETS and FuelEU make carbon-free shipping inevitable. HFO's volatile pricing and compounding penalties create an untenable cost structure.

02

Technology readiness

SMR licensing accelerating globally. Dynamic cables proven. Electric-drive ships proliferating. Every component exists today.

03

No alternative path

Every green fuel has failed to scale. Speed mandates are slowing global trade. Nuclear is the only energy source that checks every box.

The window to define the future of shipping is open. It won't stay open.

COMPETITION

The only architecture that bypasses the barriers.

COMPANY	APPROACH	PORT ACCESS	TIMELINE	SCALABLE
Allseas	Onboard SMR	Required	Long	No
Core Power	Floating NPPs	Required	Long	Partial
Maritime Fusion	Compact fusion	Required	Very long	Unproven
SMR Vendors	Near-shore float	Required	Medium	Partial
Atomarine	Offshore NPaaS	Eliminated	Medium	Yes

Atomarine is faster, cheaper, and eliminates all risk for the shipowner.

RISK INVENTORY

Four risks. Four mitigations. No PR hedges.

RISK · REGULATORY TIMELINE

Nuclear at sea — under what Flag State, on what schedule?

MITIGATION

Decouple reactor approval from vessel approval. Early Flag-State selection. Technology-taker stance — we ride the SMR vendor's certification.

RISK · TECHNICAL INTEGRATION

At-sea 100 MW HV transfer between moving ships.

MITIGATION

Dynamic HV cables already in service for offshore-wind O&M at similar voltages. Pre-Seed funds independent naval-arch review; Seed funds scaled sea-trial.

RISK · CAPITAL INTENSITY

$1B+ per LeadShip, 8+ year horizon to cash flow.

MITIGATION

Staged infrastructure capital stack. Take-or-pay capacity contracts back project-finance debt. We don't pre-fund the LeadShip at Seed — we pre-fund commitments.

RISK · PUBLIC PERCEPTION

Nuclear + ocean is a headline waiting to happen.

MITIGATION

Offshore-only operations. International waters. Flag-State jurisdiction. No port calls — ever. Narrative: 'power-sharing vessel,' not 'nuclear ship.'